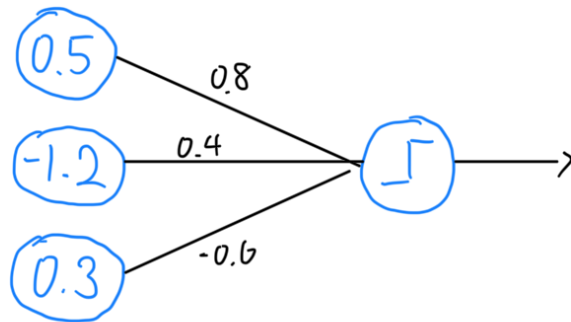


Problem A



$$(a) \quad v = 0.5 \times 0.8 + (-1.2) \times 0.4 + 0.3 \times (-0.6) = -0.26$$
$$y = 0$$

$$(b) \quad w_1^f = w_1 + \Delta w_1 = 0.8 + 0.1 \times (1 - 0) \times 0.5 = 0.85$$
$$w_2^f = w_2 + \Delta w_2 = 0.4 + 0.1 \times (1 - 0) \times -1.2 = 0.28$$
$$w_3^f = w_3 + \Delta w_3 = -0.6 + 0.1 \times (1 - 0) \times 0.3 = -0.57$$

Problem B

$$(a) \quad V_{11} = 0.3 \times 0.2 + 0.4 \times 0.2 = 0.14$$

$$\gamma_{11} = \varphi(0.14) = 0.8$$

$$V_{12} = 0.3 \times 0.2 + 0.4 \times 0.2 + 0.1 \times 0.2 = 0.16$$

$$\gamma_{12} = \varphi(0.16) = 0.83$$

$$V_{13} = 0.4 \times 0.2 + 0.1 \times 0.2 = 0.1$$

$$\gamma_{13} = \varphi(0.1) = 0.73$$

$$V_{21} = 0.8 \times 0.5 + 0.83 \times 0.5 = 0.815$$

$$\gamma_{21} = \varphi(0.815) = 1$$

$$V_{22} = 0.8 \times -0.5 + 0.83 \times 0.5 + 0.73 \times -0.5 = -0.35$$

$$\gamma_{22} = \varphi(-0.35) = 0.02$$

$$V_{31} = 1 \times 0.1 = 0.1$$

$$\gamma_{31} = \varphi(0.1) = 0.73$$

$$V_{32} = 1 \times -0.9 + 0.83 \times -0.5 + 0.02 \times 0.9 = -1.297$$

$$\gamma_{32} = \varphi(-1.297) = 0$$

$$V_{33} = 0.02 \times -0.1 + 0.73 \times 0.5 = 0.363$$

$$\gamma_{33} = \varphi(0.363) = 0.98$$

$$\gamma_{\text{pred}1} = 0.73 \times 0.6 + 0 \times -0.3 + 0.98 \times 0.1 = 0.536$$

$$\gamma_{\text{pred}2} = 0.73 \times 0.3 + 0 \times -0.9 + 0.98 \times -0.1 = 0.121$$

$$\begin{aligned}
 (b) \quad \frac{\partial E}{\partial W_{31}} &= \left(\frac{\partial E}{\partial y_{pred1}} \cdot \frac{\partial y_{pred1}}{\partial y_{31}} + \frac{\partial E}{\partial y_{pred2}} \cdot \frac{\partial y_{pred2}}{\partial y_{31}} \right) \cdot \frac{\partial y_{31}}{\partial V_{31}} \cdot \frac{\partial V_{31}}{\partial W_{31}} \\
 &= (0.536 - 0) \cdot 0.6 + (0.121 - 1) \cdot 0.3 \cdot 1.97 \cdot 1 \\
 &= \underline{0.114063}
 \end{aligned}$$

$$\begin{aligned}
 W_{31}^+ &= W_{31} - \eta \cdot \frac{\partial E}{\partial W_{31}} \\
 &= 0.1 - 0.3 \times 0.114063 = \underline{0.0657811}
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial E}{\partial W_{32}} &= \left(\frac{\partial E}{\partial y_{pred1}} \cdot \frac{\partial y_{pred1}}{\partial y_{32}} + \frac{\partial E}{\partial y_{pred2}} \cdot \frac{\partial y_{pred2}}{\partial y_{32}} \right) \cdot \frac{\partial y_{32}}{\partial V_{32}} \cdot \frac{\partial V_{32}}{\partial W_{32}} \\
 &= ((0.536 - 0) \times -0.3 + (0.121 - 1) \times -0.9) \cdot 0.1 \\
 &= \underline{0}
 \end{aligned}$$

$$W_{32}^+ = W_{32} - \eta \cdot \frac{\partial E}{\partial W_{32}} = \underline{-0.9}$$

$$\begin{aligned}
 \frac{\partial E}{\partial W_{33}} &= \left(\frac{\partial E}{\partial y_{pred1}} \cdot \frac{\partial y_{pred1}}{\partial y_{33}} + \frac{\partial E}{\partial y_{pred2}} \cdot \frac{\partial y_{pred2}}{\partial y_{33}} \right) \cdot \frac{\partial y_{33}}{\partial V_{33}} \cdot \frac{\partial V_{33}}{\partial W_{33}} \\
 &= \underline{0}
 \end{aligned}$$

$$W_{33}^+ = W_{33} - \eta \frac{\partial E}{\partial W_{33}} = \underline{0.9}$$

$$\begin{aligned}
 \frac{\partial E}{\partial W_{34}} &= \left(\frac{\partial E}{\partial y_{pred1}} \cdot \frac{\partial y_{pred1}}{\partial y_{34}} + \frac{\partial E}{\partial y_{pred2}} \cdot \frac{\partial y_{pred2}}{\partial y_{34}} \right) \cdot \frac{\partial y_{34}}{\partial V_{34}} \cdot \frac{\partial V_{34}}{\partial W_{34}} \\
 &= ((0.536 - 0) \times 0.1 + (0.121 - 1) \times -0.1) \cdot 0.2 \cdot 0.02 \\
 &= \underline{0.000566}
 \end{aligned}$$

$$\begin{aligned}
 W_{34}^+ &= W_{34} - \eta \frac{\partial E}{\partial W_{34}} \\
 &= -0.1 - 0.3 \times 0.000566 \\
 &= \underline{-0.1001698}
 \end{aligned}$$

$$\begin{aligned}
 (C) \quad \frac{\partial E}{\partial W_{21}} &= \left[\left(\frac{\partial E}{\partial \gamma_{pred1}} \cdot \frac{\partial \gamma_{pred1}}{\partial \gamma_{31}} + \frac{\partial E}{\partial \gamma_{pred2}} \cdot \frac{\partial \gamma_{pred2}}{\partial \gamma_{31}} \right) \cdot \frac{\partial \gamma_{31}}{\partial V_{31}} \right. \\
 &\quad \cdot \frac{\partial V_{31}}{\partial \gamma_{21}} + \left(\frac{\partial E}{\partial \gamma_{pred1}} \cdot \frac{\partial \gamma_{pred1}}{\partial \gamma_{32}} + \frac{\partial E}{\partial \gamma_{pred2}} \cdot \frac{\partial \gamma_{pred2}}{\partial \gamma_{32}} \right) \\
 &\quad \cdot \frac{\partial \gamma_{32}}{\partial V_{32}} \cdot \frac{\partial V_{32}}{\partial \gamma_{21}} \left. \right] \cdot \frac{\partial \gamma_{21}}{\partial V_{21}} \cdot \frac{\partial V_{21}}{\partial W_{21}} \\
 &= (0.114063 \times 0.1 + 0 \times -0.9) \times 0 \times 0.8 \\
 &= \underline{0}
 \end{aligned}$$

$$W_{21}^+ = W_{21} - \eta \frac{\partial E}{\partial W_{21}} = \underline{0.5}$$

$$\begin{aligned}
 \frac{\partial E}{\partial W_{22}} &= \left[\left(\frac{\partial E}{\partial \gamma_{pred1}} \cdot \frac{\partial \gamma_{pred1}}{\partial \gamma_{32}} + \frac{\partial E}{\partial \gamma_{pred2}} \cdot \frac{\partial \gamma_{pred2}}{\partial \gamma_{32}} \right) \cdot \frac{\partial \gamma_{32}}{\partial V_{32}} \right. \\
 &\quad \cdot \frac{\partial V_{32}}{\partial \gamma_{22}} + \left(\frac{\partial E}{\partial \gamma_{pred1}} \cdot \frac{\partial \gamma_{pred1}}{\partial \gamma_{33}} + \frac{\partial E}{\partial \gamma_{pred2}} \cdot \frac{\partial \gamma_{pred2}}{\partial \gamma_{33}} \right) \\
 &\quad \cdot \frac{\partial \gamma_{33}}{\partial V_{33}} \cdot \frac{\partial V_{33}}{\partial \gamma_{22}} \left. \right] \cdot \frac{\partial \gamma_{22}}{\partial V_{22}} \cdot \frac{\partial V_{22}}{\partial W_{22}} \\
 &= \{ [(0.536 - 0) \times -0.3 + (0.121 - 1) \times -0.9] \times 0 \times 0.9 \\
 &\quad + [(0.536 - 0) \times 0.1 + (0.121 - 1) \times -0.1] \times 0.2 \times -0.1 \} \\
 &\quad \times 0.2 \times 0.8 \\
 &= \underline{-0.0004528}
 \end{aligned}$$

$$W_{22}^+ = W_{22} - \eta \frac{\partial E}{\partial W_{22}} = \underline{-0.49986416}$$

$$\begin{aligned}
 \frac{\partial \bar{E}}{\partial W_{23}} &= \left[\left(\frac{\partial E}{\partial \gamma_{pred1}} \cdot \frac{\partial \gamma_{pred1}}{\partial \gamma_{31}} + \frac{\partial E}{\partial \gamma_{pred2}} \cdot \frac{\partial \gamma_{pred2}}{\partial \gamma_{31}} \right) \cdot \frac{\partial \gamma_{31}}{\partial V_{31}} \right. \\
 &\quad \cdot \frac{\partial V_{31}}{\partial \gamma_{21}} + \left(\frac{\partial E}{\partial \gamma_{pred1}} \cdot \frac{\partial \gamma_{pred1}}{\partial \gamma_{32}} + \frac{\partial E}{\partial \gamma_{pred2}} \cdot \frac{\partial \gamma_{pred2}}{\partial \gamma_{32}} \right) \\
 &\quad \cdot \frac{\partial \gamma_{32}}{\partial V_{32}} \cdot \frac{\partial V_{32}}{\partial \gamma_{21}} \left. \right] \cdot \frac{\partial \gamma_{21}}{\partial V_{21}} \cdot \frac{\partial V_{21}}{\partial W_{23}} \\
 &= (0.114063 \times 0.1 + 0 \times -0.9) \times 0 \times 0.83 \\
 &= \underline{0}
 \end{aligned}$$

$$W_{23}^+ = W_{23} - \eta \frac{\partial E}{\partial W_{23}} = W_{23} = \underline{0.5}$$

$$\begin{aligned}
 \frac{\partial E}{\partial W_{24}} &= \left(\frac{\partial E}{\partial \gamma_{pred1}} \cdot \frac{\partial \gamma_{pred1}}{\partial \gamma_{32}} + \frac{\partial E}{\partial \gamma_{pred2}} \cdot \frac{\partial \gamma_{pred2}}{\partial \gamma_{32}} \right) \cdot \frac{\partial \gamma_{32}}{\partial V_{32}} \\
 &\quad \cdot \frac{\partial V_{32}}{\partial \gamma_{12}} \\
 &= \underline{0}
 \end{aligned}$$

$$W_{24}^+ = W_{24} - \eta \frac{\partial E}{\partial W_{24}} = \underline{-0.5}$$

$$\begin{aligned}
 \frac{\partial E}{\partial W_{25}} &= \left[\left(\frac{\partial E}{\partial \gamma_{pred1}} \cdot \frac{\partial \gamma_{pred1}}{\partial \gamma_{33}} + \frac{\partial E}{\partial \gamma_{pred2}} \cdot \frac{\partial \gamma_{pred2}}{\partial \gamma_{33}} \right) \cdot \frac{\partial \gamma_{33}}{\partial V_{33}} \right. \\
 &\quad \cdot \frac{\partial V_{33}}{\partial \gamma_{22}} + \left(\frac{\partial E}{\partial \gamma_{pred1}} \cdot \frac{\partial \gamma_{pred1}}{\partial \gamma_{32}} + \frac{\partial E}{\partial \gamma_{pred2}} \cdot \frac{\partial \gamma_{pred2}}{\partial \gamma_{32}} \right) \\
 &\quad \cdot \frac{\partial \gamma_{32}}{\partial V_{32}} \cdot \frac{\partial V_{32}}{\partial \gamma_{22}} \left. \right] \times \frac{\partial \gamma_{22}}{\partial V_{22}} \cdot \frac{\partial V_{22}}{\partial W_{25}} \\
 &= [0 + (-0.00283)] \times 0.2 \times 0.83 \\
 &= \underline{-0.00046978}
 \end{aligned}$$

$$W_{25}^+ = W_{25} - \eta \frac{\partial E}{\partial W_{25}} = \underline{0.500140934}$$

$$\frac{\partial E}{\partial W_{26}} = \frac{\partial E}{\partial \gamma_{23}} = 0.1415 \times -0.02 \times 0.146 = \underline{-0.00041318}$$

$$W_{26}^+ = W_{26} - \eta \frac{\partial E}{\partial W_{26}} = \underline{-0.499876046}$$

$$\frac{\partial E}{\partial W_{27}} = \frac{\partial E}{\partial \gamma_{23}} = \underline{0.020659}$$

$$W_{27}^+ = W_{27} - \eta \frac{\partial E}{\partial W_{27}} = \underline{0.4938023}$$